

Patient-perceived Burden to Caregivers: Construct Components, Importance, and Wording for a Single Item

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Abstract

Background. The self-perceived burden is a patient-centred construct that refers to the perceived impact of the care recipient's illness and care needs on the physical, social, emotional, and financial well-being of their caregiver(s). The self-perceived burden may negatively impact the care recipient's quality of life and sense of dignity. It may also impact the care recipient's healthcare decision-making. The aims of this study were to 1) characterize elements that constitute caregiver burden from the perspective of patients with chronic illnesses, 2) examine the extent to which patients consider self-perceived burden relevant in their healthcare decision-making, and 3) develop a global item that captures the essential elements of self-perceived burden.

Methods. Semi-structured interviews were undertaken with US-based patients with chronic conditions. Interviews focused on the importance of self-perceived burden, components of self-perceived burden, and wording of five candidate global items encompassing the concept.

Results. The patient's self-perceived burden to their caregivers implied financial, social, emotional, and physical burdens. Patients considered self-perceived burden important to their sense of well-being and a component in their healthcare decision-making. In framing a global item, "I feel like a burden to my friends/family," was perceived by patients to encompass the construct of interest broadly.

Conclusion. US-based patients felt self-perceived burden is a highly relevant concept, and a global item may capture this distinct patient-centered construct and spillover effects to family/caregivers. A Self-perceived burden global item can inform clinical policies aimed at improving care for each patient and social care services to improve patients' and their caregivers' quality of life.

Introduction

Patients cared for by their loved ones often experience the sense that they are a burden, also referred to as "self-perceived burden" (SPB), to those caring for them (1). SPB, as a construct, may derive from the burden patients feel they have imposed on others socially, emotionally, and financially due to their health limitations (2–6). SPB is often reported in advanced diseases, ranging from 39–65% among patients receiving palliative care for cancer (7, 8), 65–70% among stroke survivors (4, 9), and over 70% among patients entering treatment for chronic nonmalignant pain (10). SPB negatively impacts patients' sense of dignity and the health and well-being of patients and their caregivers (11–13). Additionally, such feelings may impact decision-making about what treatments to accept and may play a role in a patient's decision to end their own life (8, 12, 14–18).

SPB is experienced within the relationships between patients and informal caregivers (19). Today, more than 1 in 5 Americans are informal caregivers, having provided care to an adult or child with special needs at some time in the past 12 months (20, 21) in health, social, emotional, and financial domains (22, 23). Collectively, informal caregivers provide 75–80% of total care hours for older Americans, a substantial amount of care that would otherwise have to be provided in their absence (24). A recent study

critically analyzed SPB as a concept, and concluded that SPB has psychological, relational (concern from the impact on others of one's illness and care needs), and dimensional attributes (physical, emotional, and financial) that are shaped by cultural and sociopolitical structures (25). To date, most of the research on SPB has been conducted in countries outside the US (25). Since cultures and their perception of SPB can vary worldwide (3), it is unclear if SPB is viewed differently by patients in more individualistic cultures, which shapes the predominant model of caregiving in the US (26). Qualitative aspects of SPB from diverse cultures and illness trajectories need to be explored to gain more comprehensive understandings (19).

Self-perceived burden (SPB) and caregiver burden are defining features of care relationships that have been insufficiently studied together. Additionally, SPB may be an underrecognized factor impacting the families of caregivers and offers an additional source of information in relationship to caregiver health and well-being. In a scoping review of the literature identifying measures of SPB, we identified one patient-reported outcome measure (PROM), the 10-item Self-perceived Burden Scale, with limited use and evidence of psychometric properties (2, 3, 6). However, because of concerns about the patient respondent burden, a comprehensive global item SPB may be desirable to supplement existing PROMs. In addition, a single global item would have the advantage of potential incorporation into an existing generic health and social care descriptive system for valuation. This study aims to 1) characterize elements that constitute caregiver burden from the perspective of patients with chronic illnesses, 2) examine the extent to which patients consider SPB important in their management and healthcare decision-making, and 3) develop a single global item that captures the essential elements of SPB. A clearer understanding of SPB as a concept and a global SPB item could be instrumental in providing better support services for patients and their families.

Methods

Generation of Global Self-Perceived Burden Candidate Items

We performed a scoping literature review (27, 28) to identify existing measures of SPB and how the concept was operationalized. The Self-perceived Burden Scale (SPB-Scale) (3) was the primary measure identified. In addition, based on the evidence from the literature, some concepts such as "bother" (29) and "stress" (30) (29) were identified as burden-related concepts (31). Several aspects were considered when selecting the response options (32). They included whether frequency or intensity best distinguished the level of perceived burden for the specific wording used. Level of agreement and agree/disagree were also included as options, as some might consider them simpler response options. A default position of a 5-point ordered response scale was adopted based on existing, commonly used measures (33, 34). A default recall period of 7 days was chosen to ensure enough information was collected while maintaining the accuracy and salience of responses (35). A total of 5 candidate items were drafted (Fig. 1).

Study Population and Recruitment

Our study was conducted during the face and content validity stage of a large international project, the EQ Health and Wellbeing, that aimed to develop a new generic PROM (36, 37). The University of Illinois Chicago institutional review board approved the study (#2018 – 0817). Participants for the interview were purposively recruited to capture diversity in demographic, social, and health characteristics (having acute and/or long-term physical or mental health conditions). Recruitment was done in two ways: a flyer at the University of Illinois Hospital & Health Sciences and through Research Match, and a national health volunteer registry supported by the U.S. National Institutes of Health (38). Participants were briefed on the purpose of the study and signed informed consent. Two female research assistants and co-authors (A.L.M, M.K.) with formal training in qualitative research conducted the interviews in person between October and December 2018 at the University of Illinois Chicago, College of Pharmacy. Interviews were audio-recorded, and each participant was compensated 60\$ for their time and travel cost.

Interview Guide

The interview was based on a semi-structured interview guide (Supplement 1) using cognitive debriefing (39). Prior to launch, the interview script was pre-tested with 2 graduate students and pilot-tested with the first 2 participants. First, participants were asked general questions to provide their perspective on the impact of health problems on family and friends. Questions were open-ended, supplemented with probes (40), to allow the respondent to answer the question in their own words and enable different perspectives to emerge. In the second part of the interview, we explored candidate SPB items' face and content validity to identify preferred wording and discern which item most appropriately measured the concept of interest (41, 42). Respondents were probed on written drafted items based on four areas of the cognitive response model developed by Tourangeau (39): 1) Comprehension (*i.e.*, understanding of a concept); 2) Retrieval (*i.e.*, respondent's ability to retrieve the information from memory given recall period (43)); 3) Judgment (*i.e.*, how information is combined, supplemented, and extrapolated from information that has been retrieved); and 4) Response (*i.e.*, how the respondent managed to map their desired response onto the preferred response scale (43)). In the end of the interview, respondents were asked whether they had a preference for one item and response scale. One researcher (A.L.M.) led the discussion, and the other (M.K.) took notes and provided auxiliary probes. All interviews were audio-recorded.

Analysis

Interviews were transcribed verbatim by two study team members (M.K., C.E.R.) who subsequently conducted the coding analysis using NVivo, Version 12 (QSR International) (44). Coding was conducted using thematic analysis based on the Framework Method (45, 46), which enables a combination of deductive (from *a priori* themes found in the literature) and inductive (eliciting new themes from participants) qualitative analysis (47). Coders independently reviewed 5 randomly selected transcripts before meeting to develop an initial set of codes. For codes that emerged in the patient's perspective of caregiver burden, a concept analysis framework (31, 48) was organized to explore the antecedents (events that take place before the occurrence of the concept), attributes (the meanings underlying the concept) and consequences (events that occur as a result of the caregiver burden), guided by a recently published SPB concept analysis (25). For exploring SPB candidate items, codes were aligned with

Tourangeau's response model. The coders independently coded the next set of 5 transcripts and discussed any additional need for refining codes. The remaining interviews were coded, and discrepancies were discussed and resolved by consensus. The sample size was determined by theoretical saturation (49) when interviews yielded no new information. Our study followed established practices for COnsolidated criteria for REporting Qualitative research (COREQ) in carrying out the data collection and analysis (50) (Supplement 2) and best practices for developing and validating scales for health, social, and behavioral research in developing an SPB-global item (51).

Results

Study sample

A total of 19 patients provided consent and completed interviews that averaged twenty minutes in length. One participant declined to participate in the study during recruitment when informed that the interview would be audio-recorded. Patients ages ranged from 23 to 76 years with a mean (SD) of 53.8 (13.8). The overall sample was 47.4% male. As presented in Table 1, approximately half of the respondents identified as White/Caucasian, 47.4% as Black/African American, and 10.5% as Hispanic. The majority of respondents (73.8%) had at least one chronic condition, such as severe respiratory disease, arthritis or cancer and 15.5% identified as informal caregivers.

Table 1
Participant's Demographic and Clinical Characteristics

Characteristic	N=19	%
Age		
Age, mean/SD	53.8 (13.8)	
18–39	4	21.1%
40–59	8	42.1%
60+	7	36.8%
Age range	23– 76	
Gender		
Male	9	47.4%
Female	10	52.6%
Race		
White/Caucasian	10	52.6%
Black/African American	9	47.4%
Ethnicity		
Hispanic	2	10.5%
Non-Hispanic	17	89.5%
Insurance status		
Medicare	5	26.3%
Medicaid	3	15.8%
Private insurance	8	42.1%
No health insurance	1	5.3%
Medicare and Medicaid	2	10.5%
Education		
No high school	1	5.3%
High school	4	21.1%
Bachelor's Degree	8	42.1%

Characteristic	N = 19	%
Age		
Graduate Degree	6	31.6%
Conditions		
Asthma, emphysema, or chronic bronchitis	7	36.8%
Arthritis or rheumatism	6	31.6%
Cancer diagnosed in last 3 years	3	15.8%
Diabetes	2	10.5%
Digestive problems (such as ulcer, colitis, or gallbladder disease)		0.0%
Heart trouble (such as angina, congestive heart failure, or coronary artery disease)	2	10.5%
Do you have any long-standing physical or mental impairment, illness, or disability?	14	73.7%
How difficult is it for you to meet the monthly payments on your bills?		
Not difficult at all	12	63.2%
Slightly difficult	3	15.8%
Somewhat difficult	3	15.8%
Very difficult	1	5.23%
Extremely difficult	0	0.0%
Is there anyone, either living with you or not living with you, who is sick, disabled or elderly whom you look after or give special help to?	3	15.8%
Quality of Life		
Overall, how satisfied are you with your life nowadays? Mean (SD) Range (1–10)	8	1.29
EQ-5D-5L*	0.84	0.20
EQ VAS	77.3	14.78
*US value set, based on Pickard et al. (72)		

Interview Results: Patient-perceived burden to caregivers

All respondents placed high relevancy and importance on measuring self-perceived burden. Several themes were identified within the antecedents, attributes, and consequences of the conceptual SPB framework (Fig. 2). Respondent quotes and additional details for each theme are presented in Table 2.

Table 2

Exemplary quotes and identified themes of caregiver burden from the perspective of patients

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
Lifestyle factors • Diet • Obesity • Physical activity • Prevention of chronic disease • Medication adherence • Receipt of preventive care and services • Prevention of disease complications	"There are a lot of circumstances when individuals who do not take care of themselves, they have to rely on their family to take them to store, to doctor, and they can be young. I see young people who have a stroke, and obese, I see it so often. It's indicative of a culture in the US" (Male, 62yrs). "I am very active in managing my health and I want to be active in my kids' life like everybody else. I want to be independent. I don't want to die of things I can prevent, like diabetes and hypertension. I own it. I am concerned about it; I try to do all I can and manage my health to not come down with the preventable disease. I do not want to be a burden to my friends and family" (Male, 62yrs).
Health situation • Severity of disease and/or symptoms • Disease progression and complications • Degree of disability (increase in dependency on (instrumental) activities of daily living) or frailty • Accumulation of multiple conditions • Need for emotional encouragement	"I needed a lot of care, I could not drive after chemotherapy, I needed help after chemo. When I was at home, I could not take care of cooking my food and any of that. I could not drive of my chemo. Some days I could not even get out of bed, I was nauseated and tired" (Female, 58yrs). "My mom has an issue with the esophagus, and she has to go to the Mayo clinic, so my sister has to take a day off and take her to the doctor and then wait and then take her back. It affects the whole family. Both my sisters constantly have to take her to doctor's appointments. It affects both of my sister's families. It has a great impact" (Female, 38yrs).
Financial Resources	

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
Insufficient financial resources • Savings • Retirement • Other assets • Benefits (e.g., long-term disability) • Loss of job or income • Access to other economic resources (e.g., family)	"I had a long-term disability. I had not worked in a year and was paid like I worked. A lot of people in my cancer support group do not have that. I am incredibly fortunate to have been able to actually save money when I was sick. I do not know how we expect people to take care of themselves and get better. People work in jobs that are extremely stressful and if they do not, they don't have income, and that is a horrible situation" (Female, 58yrs). "Many diseases need chronic care. You need to hire help or find some relative to do it if you cannot pay for it" (Male, 69yrs).
Healthcare system • Lack of health insurance • Accessibility of healthcare • Underinsured • Health insurance literacy	"What is driving me crazy is people are not able to keep up with the changes in healthcare policy so that sometimes they do not know they have access to things they think they do not have" (Female, 67yrs). "I take the US healthcare system is horrible. I often think about moving to Canada for many reasons. Everybody has problems of some size, but it's worse here than anywhere, and that is known fact. It is commonly known. Insurance is very big, and even if you have, it might not be good. More often than not it is not good" (Male, 69).
Sociocultural Environment	
Family support • Presence of family members that can share caregiving responsibilities • Ability of family members to support care-recipients	"We assume that everyone is self-sufficient and people who are not, they need a great deal of help. When someone is sick, that only amplifies. My sister who worked, she had to reschedule her classes to be with me for surgery and stay with me. And she took a leave of absence. Not a lot of people can do that. But I am aware many people in my cancer support group have no support, no family here, and they have to go to work" (Female, 58yrs). "I have two older sisters, of whom one is retired. She had nothing else to do and helped me out a lot. I do not know what people who do not have someone who is retired to take care of them. That is what my sister did for me" (Female, 58yrs).
Social support • Social participation • Emotional support outside family help to reduce caregiver's strain	"I think it's important to develop social relationships to augment or replace the family relationship. It is going to take off the pressure of the family, but you have to build support network around it" (Female, 64yrs).

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
Cultural norms • Patient perception on transition to care recipient as a normal process or as a self-perceived burden	“I think about what if I get older, is there family who will care for me. My cousin has her dad with Parkinson’s, and she is their primary caregiver, and I think I could not do that for my dad. She like changes his diapers, and I could not do that” (Female, 44yrs).
Relationship Quality • Influence of the relationship between care-recipient and caregiver on the patient perception of being a burden.	“Stress is there probably with a significant other more than necessarily a relative or a friend” (Female, 67yrs).
Accumulation of roles • Multiple responsibility conflict (e.g., being a parent to a child at the same time as being a caregiver to parents)	“When parents are middle-aged, they have elderly parents that they have to take care of, make decisions for and take responsibility. That can impact finances and emotions, it can cause problems between sibling who take care of the brunt of it; the people who are younger than me, in their 40ties or 50ties they are torn apart because they have no life of their own” (Male, 48yrs). “Whether there is a person who is ill is older or young as a child, and there are other children involved can influence caregiver burden” (Female, 33yrs).
ATTRIBUTES	
Financial Burden	
Direct and indirect costs of care • Medical bills • Health insurance and out of pocket costs • Medication costs • Care-related expenses • Extended care costs • Transportation costs • Time (taking time off work) • Loss of savings and/or other assets	“My health condition impacts my family because I am constantly begging them for money” (Male, 67yrs). “Even if you are in extended care on your own, then there is a frustration in trying to make sure that you are safe in your home, that you are taking your medications and that you have help in your home in whatever help you need, and that is a financial drain to your resources or your family. Even in extended care, that is expensive, and it’s a drain on your estate and everyone else” (Female, 67yrs). “Sometimes people do not have insurance, and families have to take on their financial debt. And that can be very hard on the family because everybody does not have money to pay. So that sometimes would take a toll on the family member because they are trying to find a way to help them” (Female, 23yrs).

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
Financial distress • Unpredictability of future costs • Financial sacrifices • Concerns about current and future loss of savings and/or income • Worry about being able to afford healthcare costs • Difficulty paying monthly bills	“So financially, you do not know how to plan for some of the things. Planning for the disease is stressful due to unpredictability (Female, 67yrs). “With a family member, there is emotional, the finances, the planning, where they are going to live if they need to be taken care of. So there are lots of variables that make it complicated. A financial component can lead to more emotional. It’s a cycle” (Female, 58yrs).
Emotional	
• Concerns/worries about disease trajectory • Caregivers having to face certain diseases of their loved ones	“Some health conditions, depends on how severe the problem is can cause caregiving problems that are a big deal, that are emotional problems that can be quite difficult. I’ve had to put one of my daughters in rehab, it was a big deal” (Male, 76yrs). “Emotional burden is when you are sick or a family member is sick, it is a toll on loved ones; whenever you are going through, your family is taking it emotionally because they do not know which way is going to go” (Female, 56yrs). “There is definitely emotional stress, I see it every time I see my mum; she is so emotional about everything and about how her medical problems are affecting us. And then they affect her” (Female, 38yrs).
Physical	
• Assistance with (instrumental) activities of daily living, such as dressing, help w/ getting out of bed, walking, eating, groceries, help with prescription medication, meal preparation, housework, and taking someone to the doctor	“It can be like physically having to take care of a person that is ill” (Female, 33yrs). “Burden can be physical because family have to help you” (Female, 56yrs). “I can be bedridden; they can be terribly manageable, bathing me, washing me” (Male, 62yrs).
Social	

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
Impact on family relationships • Family conflict due to disproportionate division of caregiver activities and financial costs • Change in family dynamics due to caregiving	“It’s a lot of burden on somebody, and I have seen it cause problems in the siblings and who are doing their fair share, and who is not doing their fair share, and people not agreeing financially on what they should be doing” (Female, 63yrs). “There are several siblings involved in responsibility and some are getting the brunt of everything, and the others are not. This causes the financial, emotional, and long-term relationship between siblings to form resentment” (Female, 64yrs). “There are some problems I had that I think had detrimental effects on my family. That made me feel uncomfortable” (Male, 76yrs). “When people are sick, their families face it through, for example, when they go to the hospital or in assisted living. Family is dealing with it on a daily basis” (Female, 56yrs).
Social withdrawal • Isolation from friends and family (because they feel burdened by patient’s problems).	“How this usually happens is that if friends are being distressed by my health condition, they will not be as close, so they become more isolated and get distant from a me which is an impact of stress on others” (Female, 67yrs).
CONSEQUENCES	
Financial Toxicity	

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
• Alteration of treatment decisions due to cost • Avoidance of medical care (emergent or non-emergent) • Cutback on other expenses (food, living expenses, accommodation) to cover medical bills • Job-caregiving conflict (job status change or loss, reduction of working hours)	"Our healthcare it is totally broken. I hear it very often that people have to choose between buying their medication and paying bills because of disparities and income and access to healthcare. I hear it very often. It's very common. It is something that needs to be addressed aggressively in the US" (Male, 62yrs). "Before Obamacare, before they expanded Medicare, people would not go to the doctor because they could not afford to go" (Male, 58yrs). "Financial situation can compromise patient's health, no question about it. People think about financial problems before they seek health care" (Male, 76yrs). "Financial situation is a huge issue in the US. For certain surgeries, even if I fall down the stairs and you call 911, I am not going to be able to afford an ambulance ride and stay at the hospital, even with insurance" (Female, 38yrs). "Some older people have to think about whether they are going to eat over medicine or go to the doctor. And that is a hard choice because they have to eat, and they have to go to the doctor. But if they do not have money or insurance, what are they going to do?" (Female, 56yrs). "My friend in the nursing home, she had a situation with dental care. Her financial advisor said, "you cannot consider anything until 1st of the year". You are making yourself more ill, perhaps or putting yourself in a life-threatening position" (Female, 67yrs). "I can think of one example when I know someone did not go for care because of becoming a financial burden" (Female, 63yrs). "I know a lot of people who if they can take care of themselves at home, they will do it. Other than broken bones, I knew people who glued their skin together because they did not want to get stitches. They could not afford it because they did not have insurance" (Female, 58yrs). "If I was a financial burden to my family would bother me. It would affect the decisions you make to seek help, especially if you do not have anyone to help and you have to pay out of pocket, or private insurance, or co-pay" (Male, 58yrs). "Some people do not take medicine because they cannot afford it. They don't go to the doctor until it's too late. Then they go to ER, and they in the ER say it's too late" (Male, 69yrs).
Decreased Health-related Quality of Life	

THEMES	SAMPLE QUOTES
ANTECEDENTS	
Caregiver Setting	
• Decrease in physical and mental well-being • Physical and psychological health deterioration (e.g. exhaustion) • Not having life of their own, loss of self and personal social life • Failure to address one's own problems (leads to more health deterioration)	"If you have a family member that you are caring for, often your own mental and physical health takes a back seat. I think interventions both for patient and also for the families are really important" (Female, 44yrs). "In the US healthcare system, health conditions have a huuuuuge effect on family/caregiver's daily activities" (Male, 69yrs). "I am thinking of my neighbor whole husband has Alzheimer. I am sure expenses are prohibited and her life and things she wants to do are on back burner and when I see her, she looks exhausted. That is one thing I think about, how do you take care of yourself when you have to take care of someone else. She has 3 sons, but none of them come to help. I think it causes animosity, estrangement in families, and resentment. Plus, the drain is financial and emotional" (Female, 64yrs). "When parents are middle-aged, they have elderly parents that they have to take care of, make decisions for and take responsibility. That can impact finances and emotions, it can cause problems between sibling who take care of the brunt of it; the people who are younger than me, in their 40ties or 50ties they are torn apart because they have no life of their own" (Female, 64yrs).

"It gives the sense of how they see their illness and relation to their family and what their home situation is like." (Female, 56yrs).

Antecedents

In antecedents respondents discussed caregiving setting, financial, and sociocultural factors as directly relating to care needs. Great concern was expressed for the lack of financial resources and adequate health coverage, which is complicated by changes in health policy and uncertain access to needed services. The sociocultural environment in which the care-recipient and caregiver play mediating roles was also discussed. This included having family support to divide caregiving responsibilities and avoid an accumulation of roles (e.g. being a parent and a caregiver at the same time), along with a social network outside of the immediate family for emotional support. Some respondents noted that cultural upbringing impacts what is expected from them and that relationship quality between care-recipient and caregiver dictates the caregiver burden and its appraisal.

"We assume that everyone is self-sufficient and people who are not, they need a great deal of help. When someone is sick, that only amplifies. My sister who worked, she had to reschedule her classes to be with me for surgery and stay with me. And she took a leave of absence. Not a lot of people can do that. But I

am aware many people in my cancer support group have no support, no family here, and they have to go to work” (Female, 58yrs).

Attributes

The attributes included financial, emotional, physical, and social themes, which were often discussed as interrelated. All respondents emphasized financial burden, which can be both objective and financial distress (52, 53). Objective financial burden consists of care-related expenses that often impact the families' financial resources. These objective sources of burden are appraised by individuals, which may then manifest as financial distress with concern about the current and future financial situation and inability to plan financially due to unpredictable disease trajectory. In addition, there is an emotional burden: the worry caregivers experience about their loved ones and the uncertainty of future disease. Physical burden relates to helping with (instrumental) activities of daily living. Finally, changes in family dynamics can impact family relationships and lead to conflicts about the uneven division of caregiving activities and financial cost.

“With a family member, there is emotional, the finances, the planning, where they are going to live if they need to be taken care of. So there are lots of variables that make it complicated. A financial component can lead to more emotional. It’s a cycle” (Female, 58yrs).

Consequences

Respondents mentioned many examples they witnessed among their family and friends, such as avoiding or delaying seeking medical care when needed, budget constraints on essential living expenses due to care expenses, and even having to choose between eating and paying for medication. This is further aggravated by difficulty in maintaining outside employment due to an illness in both care recipients and caregivers. This strain has an impact on caregivers' physical and mental well-being, which can lead to not having a life of their own and health deterioration.

“That is one thing I think about, how do you take care of yourself when you have to take care of someone else” (Female 64yrs).

Finally, respondents recognized that the caregiver burden is a dynamic process that depends on individual disease trajectory. When faced with stressful or burdensome situations, such as financial problems during serious illness, one's ability to cope with the situation is an essential factor that impacts subjective perceptions. According to Theory of Stress and Coping (54, 55), external stressors are appraised by individuals within the context of their own history, resources, problem-solving skills, and emotional resilience.

Item-specific findings: face and content validity

Amongst the items, the wording “I feel like a burden to my family” with a severity response scale encompassed the issue/construct of interest. Burden implied financial, social, physical, and emotional burden, without the need for further clarification of the concept. However, respondents found the inclusion

of examples useful to prompt different ideas of burden and get a sense of whether they feel they are burdening. As one respondent noted when explaining her item preference: *“burden rounds it up, everything”* (Female, 56yrs). Among response options, severity was most preferred, and frequency second preferred. A summary of the core findings and any validity issues for each item is outlined below and summarized in Table 3.

Table 3
Self-perceived burden candidate item's face and content validity

	Summary of Face Validity	Face Validity Sample Quotes	Summary of Content Validity	Content Validity Sample Quotes
I am a bother to my family.	Comprehension: • Ambiguity in meaning (whether bother here relates to health) • Vague meaning (mostly unrelated to health) • False inference (people may not see themselves as a bother)	"Is this questionnaire only about healthcare? You could be a bother for a lot of reasons." "It can be vague, but it can mean anything, that I am able to live to my responsibilities, maybe something of health situation when I am not able to work or you know, capable, bedridden, it could be anything. So vague."	Bother was interpreted as: "Annoying, "casual annoyance," "irritating," "something you deal with," "not much of a problem," "nuisance to my family," "being an obligation," "being extra problem for your family," "inconvenience," "causing your family and friends to get tired of dealing with your problems"	"Bother means when people are annoyed, irritated with you." "I don't think bother is the right word, it sounds like annoying." "Bother is one thing, the burden is another. The burden is more financial, as well as time per caregiving. A bother is I think from the patient's point of view, they feel of that, that is upsetting to them, more occasional. Where sometimes, occasional bother is not going to be hard on family and friends. But it can quickly turn into a burden. Bother is a step lower than a burden."

	Summary of Face Validity	Face Validity Sample Quotes	Summary of Content Validity	Content Validity Sample Quotes
I am a burden to my family.	Comprehension: Ambiguity in meaning; respondents recommended to provide some examples to give people options on which burden they may be and they "get a sense if they feel they are burdening at all."	"I find some people might not find it easy to answer because people do not always see themselves as other people see them." "It is going to be uncomfortable for some people, but at the same time it is going to make them look at themselves."	Burden was interpreted as: • Financial (paying for hospitalization, medication, professional care) • Time per caregiving (having to take time off work, or transportation, drive someone somewhere) • Physical (helping with (instrumental) activities of daily living such as washing, eating, help around the home, helping physically to take someone to doctor) • Emotional and mental burden • Frustration (by inability powerless to help) • Stress (of losing someone) • Energy • Sacrifice of the family	"Very similar to bother, but burden means financial, emotional and time, but you are not really worth you are causing more negative feelings than positive feelings. The burden is more serious than bother. You can be burden financially, emotionally, time and sacrifice to the rest of the family" "Burden would be I am a big negative in their life. And it brings in more financial side, that bother did not bring. Burden is financial, time, energy." "Burden means physically take my mum, or financially that we happen to pay for her hospitalization." "You could be an emotional burden, a time burden, I worry about you, you are taking all my time, but you might have money, so you are not a financial burden to me, but you would think maybe you need help. You could also be a financial burden."

	Summary of Face Validity	Face Validity Sample Quotes	Summary of Content Validity	Content Validity Sample Quotes
My health problems/state causes stress to my family.	Comprehension: Ambiguous interpretation, vague meaning. Some participants thought it is not helpful to ask and will provide no information.	"I'd recommend putting in parenthesis because stress can mean different things to people." "You can be stressed about your problems, without being a burden."	Stress was interpreted as: • Emotional • Worry about health problems by family members • Pain and suffering • Pressure to take care of a family member • Emotions "like anxiety" • Financial, physical and time (in one respondent)	"Um, you're kind of leaning the door on interpretation. If I would be thinking about it, if I had three bother, is kind of annoyance, stress is a little bit ongoing, they are more worried about it, it is more emotional and the burden is the thing nobody wants to be." "Stress is mostly emotional to me at least. Burden could also be emotional but could be all those other things, but stress is focusing more on emotions. Like anxiety."
I am a financial burden to my family.	Some respondents felt it might be uncomfortable to explicitly ask (only) about financial concerns, that some might find it offensive, intrusive, or a sensitive topic. The point was also raised on comprehension and not understanding the purpose of the question.	"I think it might be a little bit causing stress to ask people that. It is not worth asking that. I do not understand why you would ask that, why you need information that for. That might make people angry or uncomfortable." "I think if you are going to do an intervention, including the financial component is important, but I do think just word burden captures the financial burden."	Financial burden was interpreted as cost related to medical services and financial distress. It is only a subset of burden and does not capture the whole concept of interest.	"You could be a financial burden, but that is only a subset of burden."

	Summary of Face Validity	Face Validity Sample Quotes	Summary of Content Validity	Content Validity Sample Quotes
Are you concerned about the burden your health imposes on your family and friends?	Some respondents felt it might be uncomfortable for some, even aggressive. Having statements is preferred for consistency in framing with other questions.	"I am uncomfortable with this question. Because it implies, you are telling them they are a burden. I feel like I am a burden is a better way. " Asking do you identify, concern as a burden, it feels a little aggressive."	Concern as a burden was interpreted more subjectively.	"It is the same thing as previous items. The difference is it's more how I am feeling as opposed to more like the previous two were more concrete."

"Bother" and "stress" were not interpreted as directly representative of the original construct of burden. Most respondents interpreted bother as unrelated to health issues, even though it was within a health questionnaire. There was ambiguity and differences in interpretation of "bother" with the most common words to describe it being "annoyance," irritated," "inconvenience,". When further probed with a question on how it differs from the concept "burden," it was considered a "step lower" than burden. Additionally, one respondent noted that burden brings a financial component, whereas bother does not imply this. Most respondents interpreted stress as having emotional implications, and the term might be too vague if we are interested in stress as a multidimensional concept with financial and physical components.

"If I had three concepts: bother, stress, and burden. Bother, is kind of annoyance, stress is a little bit ongoing, they are more worried about it, it is more emotional and the burden is the thing nobody wants to be" (Female, 63yrs).

Discussion

Using a qualitative approach, this study generated evidence for a greater understanding of caregiver burden from the perspective of patients with chronic illnesses in the US. Based on methods that drew upon current good practices for measure development (51), we were also able to identify preferred wording for the SPB-Global item: *"I feel like a burden (physical, emotional, social, and financial)."* We found that respondents described several elements relevant to SPB, such as financial, emotional, physical, and social issues. In addition, all participants viewed the "burden of their health on friends and families" as critical to their own sense of well-being and an important consideration in healthcare decision-making. Respondents often referred to their own experiences to illustrate the magnitude of the issue or the situation they observed of people to whom they were close.

Generally, our findings extend those of Cousineau et al., suggesting SPB has physical, emotional, social, and financial components (2). As a concept, patient-perceived caregiver burden is consistent with how caregivers perceive their burden, which raises the possibility of using SPB as a proxy for caregiver burden (56, 57). To be clear, we are not suggesting SPB is the same construct as caregiver-reported burden, but the two may be closely related and the extent to which SPB could substitute for caregiver-reported burden is highly relevant as it would only require a single respondent. However, the literature on the relationship between these concepts is very limited and future research should focus on quantifying the agreement between caregiver burden from the perspective of caregivers and from the perspective of care recipients.

The newly constructed SPB-Global item presents an opportunity to capture SPB as 1) a *de novo* measure and 2) as part of a health and social care descriptive system as a proxy to capture the effect of illness on family members. As a *de novo* instrument, the SPB-Global item offers an advantage to measure the SPB concept while imposing a minimum burden on respondents and is more time-efficient than the currently used multiple item SPB-Scale.

By employing this parsimonious measure of SPB, clinicians and other professionals working with care recipients and their caregivers could determine if a further assessment was needed while minimizing the time, energy, and resources required. Although SPB is a common experience in patient populations (7, 9, 10), family members and healthcare practitioners often fail to notice this problem in their patients (58–60). Such an understanding is crucial to inform clinical policies based on the moral duty to provide all patients with humane care (61). As the population ages and people with chronic diseases are living longer, a growing dependence on care is likely to result in an increasing shift from professional to informal care (62). Therefore, healthcare professionals should be aware of this concern and be prepared to reduce this sense of burden.

The person living with illness, disease, or disability may not have clear communication with their informal caregivers who support them through their medical conditions (63). The SPB-Global item can also facilitate dialogue between patients and their family members to help them discover solutions to reduce their SPB and find positive meanings in the caregiving-receiving situation. Based on a more comprehensive understanding of the SPB, health and social care can provide better support services to patients and their families.

As part of health and social care descriptive systems, the SPB-Global item could facilitate valuation as a supplemental item to an existing descriptive system for health/well-being. For example, as a “bolt-on” to the EQ-5D, it may provide an alternative means of obtaining information on caregiver health and well-being using a single source – the patient. This offers an opportunity to proxy measure the effects of illness on a patient’s family members and as a part of cost-effectiveness analyses (CEA). Several challenges are associated with quantifying the uncertainty in measuring caregiver effects and incorporating them as utilities. These challenges include determining which caregivers are affected by an intervention, quantifying and finding the suitable method to measure these “*spillover effects*”, and finally, how to aggregate and present QALYs for caregivers and patients while avoiding double-counting (64–68).

As international guidelines on conducting economic evaluations around the world are trying to find an optimal methodology to estimate and incorporate informal caregiver effects into an economic evaluation, our SPB-Global item poses two potential methodological advantages: 1) it could avoid double-counting of benefits by incorporating patient and proxy caregiver utilities in the same Quality-Adjusted Life Year (QALY) scale and 2) by asking care-recipients about the “burden to their family,” it offers the potential to incorporate the effect of multiple caregivers within the care recipient network. Future research should include psychometric testing of the proposed item alongside other instruments to inform measurement characteristics and the contexts in which the item has the potential to inform SPB (for example, alongside generic measures of health and social care).

The limitations of our study are common to qualitative research: limited generalizability, and the potential for researcher bias. All of our participants were recruited in the Chicago, IL, US area. Therefore, the study results may reflect only the views of respondents who agreed to be interviewed, and the extent to which these results generalize is uncertain. Due to the mode of administration, interviewers can potentially socially influence participants by their behavior, personal appearance, or voice characteristics, resulting in social desirability bias (49). Despite limitations, this is the first study in the US to gain a broader understanding of SPB. The strength of our study includes using cognitive debriefing as a method of interviews. Cognitive debriefing allowed participants to provide an in-depth understanding of how SPB-Global item operates, the kind of phenomena it captures, and how it ultimately serves the scientific goal (69, 70). Furthermore, it also offers possibilities for identifying threats to response error (49, 70, 71).

Conclusion

Outside the clinical setting, SPB has been an underrecognized source of insight into the impact of illness on families and may provide an alternative means of obtaining information on caregiver health and well-being in outcomes research by using a single source – the patient. We found that the SPB-Global item encompasses the construct of interest and implied financial, social, physical, and emotional burden of the respondent, aligned with how caregiver burden is operationalized. The SPB-Global item could be considered as part of a *de novo* instrument or as part of a health and social care descriptive system, depending on the goal/application. The results of this study have the potential to improve measuring SPB and, ultimately, may be utilized to improve the relationships and quality of life between care recipients and their caregivers.

Declarations

Ethics approval and consent to participate

The University of Illinois Chicago institutional review board approved the study (#2018-0817). All participants signed informed consent.

Availability of data and materials

Deidentified qualitative data transcripts and the codebook generated and analyzed during the study are available upon request to the corresponding author.

Competing interests

ASP and BM are members of the EuroQol Group. Other authors declare that they have no competing interests.

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Authors' contributions

MK, ASP and ALM contributed to the conceptualization and study design. MK and ALM were responsible for acquiring the data from patients that MK and CER analyzed. MK drafted the manuscript. All authors substantially contributed to the data interpretation, critical revision of the manuscript and agreed to its final content.

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Figures

For each of the following statements, please check one box that best describes your thoughts, feelings, and activities over the last 7 day.					
Items	Response options*				
	Not at all	A little bit	Somewhat	Quite a bit	Very much
	None of the time	Only occasionally	Some of the time / Sometimes	Often	Most or all of the time
	No difficulty	Slight difficulty	Some difficulty	A lot of difficulty	Unable
	Strongly disagree	Disagree	Neither Agree or Disagree	Agree	Strongly Agree
I feel like a <u>bother</u> to my family					
I feel like a <u>burden</u> to my family					
My health problems/state causes <u>stress</u> to my family					
I am a <u>financial burden</u> to my family					
Are you concerned about the burden your health imposes on your family and friends?					

Note: * Response options (in order of appearance in the table): Severity, Frequency, Level of Difficulty, Agree/Disagree.

Figure 1

Self-perceived burden candidate items and response option sets

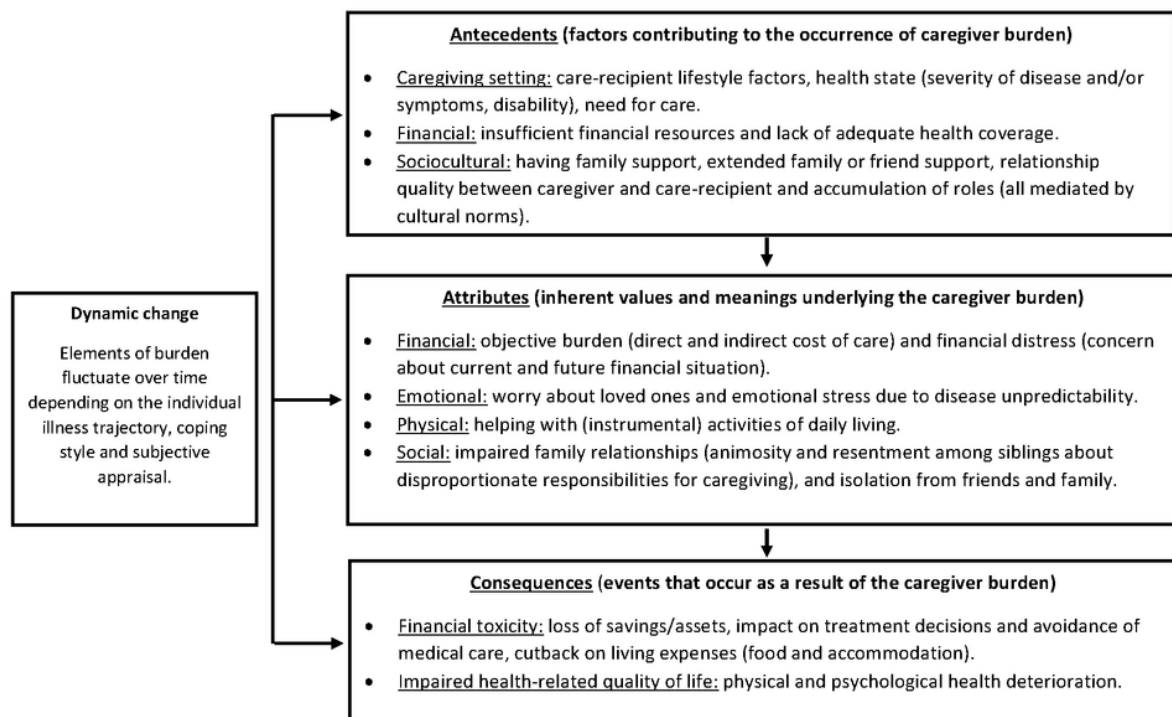


Figure 2

Conceptual components of patient-perceived burden to caregivers

Supplementary Files

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- [SupplementaryMaterialPatientperceivedBurdentoCaregivers.docx](#)